

## 교육용 자료 안내

본 자료는 Darun Color Institute의 색채 교육 과정을 위해 제공되는 교육용 문서입니다. 학생들이 안전한 재료 취급을 학습하기 위해 사용된 원료의 물리화학적 특성과 안전 정보를 공개합니다.

다만, 이 자료에 포함된 정보는 교육 목적으로만 활용되어야 하며, 상업적 목적의 제품 제조 및 판매는 엄격히 금지됩니다.

화장품 제조업 허가를 취득한 자가 독립적으로 제품을 개발·판매할 경우, 다음을 반드시 준수해야 합니다:

- 신뢰할 수 있는 화장품 원료 전문 제조사로부터 원료를 직접 확보
- 해당 제조사로부터 최신 MSDS 및 시험성적서(CoA, 안정성 시험 등)를 개별 요청
- 식품의약품안전처 규정에 따른 원료 검증 및 제품 신고

이는 소비자 안전과 법적 책임을 보장하기 위한 필수 절차입니다.

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## Educational Material Notice

This document is provided by Darun Color Institute as an educational resource for our color science training programs. To support student learning in safe material handling practices, we disclose the physical, chemical, and safety characteristics of the raw materials used in our formulations.

However, all information contained in this material is strictly for educational purposes only. Any commercial use for product manufacturing or sales is strictly prohibited.

Should you obtain cosmetic manufacturing authorization and independently develop products for commercial purposes, you must:

- Source raw materials directly from reliable cosmetic ingredient manufacturers
- Request current MSDS and certificates of analysis (CoA) individually from the respective manufacturers
- Verify all raw materials and complete product registration in compliance with Korea Food and Drug Administration (KFDA) regulations

These procedures are mandatory requirements to ensure consumer safety and fulfill legal responsibilities.

## Material Safety Data Sheet (MSDS)

### 1. Chemical Product and Company Identification

| Item                     | Information               |
|--------------------------|---------------------------|
| Product Name             | Educational Color Z-Brown |
| Recommended Use          | Cosmetic raw material     |
| Means of Identification  | Not available             |
| Relevant identified uses | Not applicable            |

### 2. Hazards Identification

| Category                 | Details                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GHS Classification       | Specific Target Organ Toxicity (Single Exposure) - Category 3 (Respiratory irritation)                                                                                                                                                                                                                                                                                                                                          |
| Hazard Statements        | H335: May cause respiratory irritation                                                                                                                                                                                                                                                                                                                                                                                          |
| Precautionary Statements | P261: Avoid inhalation of dust, fumes, gas, mist, vapor, spray<br>P271: Use only outdoors or in well-ventilated areas<br>P304+P340: If inhaled: Move to fresh air and rest in comfortable position<br>P312: Seek medical advice if you feel unwell<br>P403+P233: Store in well-ventilated place. Keep container tightly closed<br>P405: Store in locked storage<br>P501: Dispose of contents/container according to regulations |

### 3. Composition/Information on Ingredients

*\*Composition ratios are provided as approximate ranges for educational purposes*

Substance/mixture : Mixture

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment, are PBTs or vPvBs or have been assigned a workplace exposure limit and hence require reporting in this section.

### 4. First Aid Measures

| Route of Exposure | First Aid Procedure                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye Contact       | Rinse with water for at least 15-20 minutes. Seek medical attention if irritation persists.                                                           |
| Skin Contact      | Wash with soap and water for at least 20 minutes. Remove contaminated clothing and shoes before reuse. Seek medical attention if irritation persists. |
| Inhalation        | Move to fresh air immediately. If breathing is difficult, seek medical attention. If coughing or other symptoms occur, seek medical                   |

|                   |                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------|
|                   | examination.                                                                                          |
| Ingestion         | Rinse mouth with water. Do not induce vomiting. Seek medical attention immediately for large amounts. |
| Medical Attention | Ensure medical personnel are aware of substance exposure and take appropriate protective measures.    |

## 5. Fire-Fighting Measures

| Item                         | Information                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Dry powder, carbon dioxide, water spray, alcohol foam                                             |
| Specific Hazards             | Container may explode when heated. Irritating and toxic gases may be generated during combustion. |
| Special Protective Equipment | Move container from fire area if safe. Cool containers with water spray even after extinguishing. |
| Unsuitable Media             | High-pressure water streams                                                                       |

## 6. Accidental Release Measures

| Action                   | Description                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prevention               | Remove all ignition sources. Prevent dust formation. Ventilate contaminated area.                                                                                                        |
| Handling Spills          | For small spills: Absorb with sand or non-combustible material and place in container<br>For large spills: Create barriers to prevent spreading. Use clean shovel to place in container. |
| Environmental Protection | Prevent entry into waterways, sewers, basements, and confined spaces. For powder spills, cover with plastic sheet to prevent diffusion.                                                  |

## 7. Handling and Storage

| Category | Requirements                                                                                                                                      |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling | Handle carefully following safety procedures. Use full or local exhaust ventilation. Wash thoroughly after handling. Avoid inhalation.            |
| Storage  | Store in cool, dry, and well-ventilated place. Keep container tightly closed. Store away from heat, ignition sources, and incompatible materials. |

## 8. Exposure Controls/Personal Protection

| Control Type         | Requirement                                                                              |
|----------------------|------------------------------------------------------------------------------------------|
| Engineering Controls | Use local exhaust ventilation or process isolation to maintain air below exposure limits |

|                        |                                                                                        |
|------------------------|----------------------------------------------------------------------------------------|
| Respiratory Protection | Use KOSHA-certified respiratory protection appropriate for particulate matter exposure |
| Eye Protection         | Wear safety glasses or goggles. Install emergency eyewash facilities near work area    |
| Hand Protection        | Wear appropriate chemical-resistant protective gloves                                  |
| Body Protection        | Wear appropriate chemical-resistant protective clothing                                |
| Facilities             | Install emergency eyewash and shower facilities in work area                           |

## 9. Physical and Chemical Properties

| Property         | Value                                         |
|------------------|-----------------------------------------------|
| Appearance       | Cream to paste-like, brown colored semi-solid |
| Odor             | Slight characteristic odor                    |
| Viscosity        | No data available                             |
| pH               | No data available                             |
| Solubility       | Partially soluble; water component soluble    |
| Specific Gravity | No data available                             |

## 10. Stability and Reactivity

| Item                    | Information                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------|
| Chemical Stability      | Stable at room temperature and normal pressure                                                    |
| Conditions to Avoid     | Heat, sparks, flames, and other ignition sources                                                  |
| Incompatible Materials  | Combustible materials, oxidizing agents                                                           |
| Hazardous Decomposition | Container may explode when heated. Irritating and toxic gases may be generated during combustion. |

## 11. Toxicological Information

| Effect                                   | Description                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------------------|
| Acute Toxicity (Oral)                    | LD50 > 5000 mg/kg (Rat) - Iron Oxide component                                                  |
| Acute Toxicity (Inhalation)              | Dust LC50: 5.05 mg/L 4hr (Rat) - Iron Oxide component                                           |
| Skin Irritation                          | Minimal irritation potential                                                                    |
| Eye Irritation                           | Minimal irritation potential                                                                    |
| Respiratory Irritation (Single Exposure) | May cause respiratory irritation from iron oxide dust component                                 |
| Long-term Exposure Effects               | Prolonged exposure to iron oxide dust may cause respiratory effects                             |
| Carcinogenicity                          | Iron Oxide Group A4 (ACGIH) - Suspected human carcinogen (inhalation hazard for fine particles) |
| Reproductive Toxicity                    | No data available                                                                               |

|                   |                   |
|-------------------|-------------------|
| Aspiration Hazard | No data available |
|-------------------|-------------------|

## 12. Ecological Information

| Category                      | Information                                 |
|-------------------------------|---------------------------------------------|
| Fish Toxicity                 | LC96hr ≥ 50000 mg/L (Iron Oxide component)  |
| Crustacean Toxicity           | EC50 > 100 mg/L 48hr (Iron Oxide component) |
| Bird Toxicity                 | No data available                           |
| Persistency and Degradability | No data available                           |
| Bio-condensability            | No data available                           |
| Soil Mobility                 | No data available                           |

## 13. Disposal Considerations

| Item                | Information                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Disposal Method     | Dispose of contents and container in accordance with local waste management regulations                                        |
| Special Precautions | For iron oxide component: Neutralization, hydrolysis, oxidation, reduction, incineration, or solidification may be appropriate |

## 14. Transport Information

| Item                   | Information                               |
|------------------------|-------------------------------------------|
| UN Number              | Not classified as hazardous for transport |
| Proper Shipping Name   | Not applicable                            |
| Transport Hazard Class | Not applicable                            |
| Packing Group          | Not applicable                            |
| Marine Pollutant       | Not applicable                            |

## 15. Regulatory Information

| Item             | Information                                                                                    |
|------------------|------------------------------------------------------------------------------------------------|
| Regulatory Basis | Prepared in accordance with Korean Occupational Safety and Health Act and relevant regulations |

## 16. Other Information

| Item       | Information                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Disclaimer | The information contained in this document is based on accurate data. However, no guarantee is provided regarding the accuracy of results or |

|               |                                                                                                                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | values obtained when using this material for specific purposes or mixing with other substances. The company assumes no responsibility for any personal losses or property damage to buyers or third parties. |
| Prepared by   | Darun Color Institute                                                                                                                                                                                        |
| Document Type | For Educational Purposes Only                                                                                                                                                                                |